

Metrics. Tokenizer: reconstruction PSNR/SSIM on held-out frames. Action model: NMI_{adj} between action codes and VLM-judged action groups, all groups, on the frozen action-group set; same protocol for every model, random ≈ 0 . Dynamics: one-step next-frame prediction from real context on the held-out split (Push-T: NanoWM protocol, 3 autoregressive steps). – = not relevant or not measured.

Zelda (128 px)

Dynamics: 890 held-out windows, 3 context frames $\rightarrow$ next frame, decoded with context, 1 MaskGIT pass / 1 Euler step. Tokenizer recon: 220 held-out 7-frame windows. LPIPS marked * is at 4 Euler steps.

Tokenizer			Action model		Dynamics				PR	Notes
	PSNR↑	SSIM↑		NMI _{adj} ↑		PSNR↑	SSIM↑	LPIPS↓		
–	–	–	–	–	copy last frame	27.43	0.681	0.098	–	baseline
FSQ temporal	32.42	0.933	LAM (v3)	0.110	MaskGIT	26.72	0.759	–	#4, #14	original Genie pipeline
FSQ temporal	32.42	0.933	CoMo-ViT, 16 codes	0.561	MaskGIT	28.02	0.817	0.096		#20
FSQ temporal	32.42	0.933	CoMo-ViT, 16 codes	0.561	flow matching	28.34	0.816	0.099*	#19	FM > MaskGIT twin
FSQ per-frame	32.12	0.929	CoMo-Tok, 16 codes	–	MaskGIT	27.39	0.789	0.109	#21	CoMo on our tokenizer features
FSQ per-frame	32.12	0.929	CoMo-Tok, full z	–	MaskGIT	29.06	0.865	0.085	#21	
FSQ temporal	32.42	0.933	CoMo-ViT, full z	0.561	flow matching	29.81	0.884	0.086	#19	
continuous temporal KL-VAE	44.46	0.995	CoMo-ViT, full z	0.561	flow matching	33.78	0.937	0.036	#24	+4 dB from continuous tokens
continuous temporal KL-VAE	44.46	0.995	CoMo-ViT, full z	0.561	FlowC2S	34.09	0.940	0.034	#25	fine-tuned from row above
continuous temporal KL-VAE	44.46	0.995	CoMo-ViT, full z	0.561	Bi-flow	33.72	0.933	0.038	#25	from scratch, 30k
–	–	–	CoMo-ViT, full z	0.561	CoMo decoder	33.41	0.931	0.038	#20	CoMo decoder as dynamics
–	–	–	CoMo-ViT, 16 codes	0.561	CoMo decoder	30.15	0.844	0.050	#20	
Merged training stages — in progress										

Push-T (128 px, ground-truth actions)

NanoWM protocol: 21 val episodes, 256 clips, 1 real frame → 3 predicted autoregressively. NanoWM predictions re-scored with our scorer (validated against theirs).

Tokenizer			Action model		Dynamics				PR	Notes
PSNR↑	SSIM↑		NMI _{adj} ↑		PSNR↑	SSIM↑	LPIPS↓			
–	–	–	–	–	copy last frame	26.59	0.950	0.056	–	baseline
–	–	–	ground truth	–	NanoWM-B/2 (100k)	35.10	0.985	0.009	–	external reference
FSQ temporal	42.84	0.996	ground truth	–	MaskGIT, 30k	34.54	0.983	0.018	#8	FID 26.0 vs 9.0

Pong (64 px)

Held-out split, 1340 windows, 3 context frames → next frame, 10 MaskGIT passes. Final checkpoint.

Tokenizer			Action model		Dynamics				PR	Notes
	PSNR↑	SSIM↑		NMI _{adj} ↑		PSNR↑	SSIM↑	LPIPS↓		
–	–	–	–	–	copy last frame	30.06	0.953	–	–	baseline
FSQ temporal	45.70	0.997	LAM (16 codes)	0.416	MaskGIT, 39k	42.39	0.993	–	#5	
FSQ temporal	45.70	0.997	random code	–	MaskGIT, 39k	35.95	0.978	–	#5	same model, random action